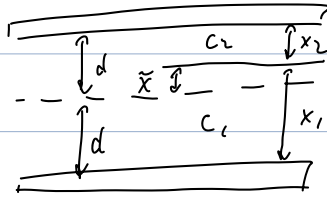


Capacitive Acc



$$\epsilon_A: \epsilon_0 \epsilon_r A, \quad C_0 = \epsilon_A \frac{1}{d}$$

$$C_1 = \epsilon_A \frac{1}{x_1} = \epsilon_A \frac{1}{d+x} = \frac{1}{\frac{1}{\epsilon_A d} + \frac{1}{\epsilon_A x}} = \frac{1}{C_0 + \Delta C} \approx C_0 - \Delta C$$

$$C_2 = \epsilon_A \frac{1}{x_2} = \epsilon_A \frac{1}{d-x} = C_0 + \Delta C$$

$$C_2 - C_1 = 2\Delta C = 2\epsilon_A \frac{x}{d^2 - x^2}$$

$$\Rightarrow \underbrace{\Delta C x^2}_{\text{negligible}} + \epsilon_A x - \Delta C d^2 = 0$$

$$\Rightarrow x \approx \frac{d^2}{\epsilon_A} \Delta C = d \cdot \frac{\Delta C}{C_0}$$

now, we can use the formula for mech acc:

$$a = \frac{k}{m_p} x = \frac{k d}{m_p C_0} \Delta C$$